

Jakub Kopal

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Computational neuroscientist translating machine learning research into clinical and applied practice.

Working experience

- 2024 – **University of Oslo, Oslo, Norway**, *MSCA Postdoctoral fellow*,
Present *Supervisor: Ole A. Andreassen*,
Marie Curie Postdoctoral fellowship
Coordinate AI research projects between University of Oslo and UC San Diego (co-supervisor: Anders Dale), translating predictive models for treatment response into clinical research settings. Participate in Horizon Europe projects (environMENTAL, GenAI4EU, EU4Health - Brain AID). Contribute to grant proposals for Horizon Europe, focusing on AI for healthcare. Deliver analytical workflows for large-scale health datasets.
- 2021 – 2024 **McGill University & Mila - Quebec AI Institute, Montréal, Canada**, *Postdoctoral fellow*,
Supervisor: Danilo Bzdok, *Co-supervisor: Sébastien Jacquemont*,
Mentored undergraduate and graduate students
Applied advanced machine learning methods to large-scale biomedical datasets.
Led collaborations between Mila and CHU Sainte-Justine, translating clinical research questions into machine-learning workflows.
- 2017 – 2021 **Czech Academy of Sciences, Prague, Czech Republic**, *Ph.D. student*,
Institute of Computer Science, Complex networks and brain dynamics group,
Developed computational models and statistical tools for analyzing brain connectivity .
Applied advanced statistical modeling and network theory to neuroimaging data.
Presented findings to clinical audiences without technical backgrounds.
- Jun 2015 – **University of Glasgow, Glasgow, UK**, *Internship*,
Aug 2015 Department of Bioengineering,
Project: Development of a rapid point of care diagnostic device utilizing surface acoustic waves.

Education

- 2017 – 2021 **Brain and Cognition Research Center (CerCo), Toulouse, France**,
Doctoral studies in Neuroscience (cotutelle PhD with UCT Prague),
Supervisor: Emmanuel J. Barbeau, Ph.D.
- 2016 – 2021 **University of Chemistry and Technology, Prague, Czech Republic**,
Doctoral studies in Technical cybernetics (cotutelle PhD with UPS Toulouse),
Research on computational modeling, signal processing, and machine learning in biomedicine.
Teaching: Applications of Computer Science (Czech and English language),
Supervisor: Oldrich Vysata M.D., Ph.D., Co-supervisor: Jaroslav Hlinka, Ph.D.

- 2013 – 2016 **University of Chemistry and Technology, Prague, Czech Republic,**
Master's degree,
 Field of study: Applied engineering informatics,
 Focus on optimization methods, process modeling, and applied machine learning.
 Relevant coursework: Engineering Optimization, Numerical Methods, Mathematical Modeling of Processes, Neural Networks, Pattern Recognition, Discrete Event Simulation.
- Sep 2014 – **University of Oulu, Oulu, Finland,**
 Dec 2014 *Erasmus exchange student,*
 Field of study: Bioengineering and computer science,
 Project: Dynamic measurements of balance with footwear sensors for BoneWell Intelligence Ltd.
- 2010 – 2013 **University of Chemistry and Technology, Prague, Czech Republic,**
Bachelor's degree,
 Field of study: Process engineering, informatics, management,
 Strong foundation in chemical engineering and control systems.
 Relevant coursework: Measuring and Control Engineering, Chemical Engineering, Signal Processing, Enterprise Process Management, Mathematics, Physics.
Bachelor thesis: Dynamic Parameters of EEG during Meditation.

Academic record

Google scholar <https://scholar.google.com/citations?user=GfLE4fgAAAAJ>.

Selected publications

The end game: Respecting major sources of population diversity,
 Kopal, J., Uddin, L., Bzdok, D.,
 Nature Methods.

A pattern-learning algorithm associates copy number variations with brain structure and behavioural variables in an adolescent population cohort,
 Kopal, J., Jacquemont, S., Bzdok, D., et al.,
 Nature Biomedical Engineering.

Rare CNVs and phenome-wide profiling highlight brain-structural divergence and phenotypical convergence,
 Kopal, J., Jacquemont, S., Bzdok, D., et al.,
 Nature Human Behavior.

Deep learning reveals that multidimensional social status drives population variation in 11,875 US participant cohort,
 Marotta, J., Kopal, J., Bzdok, D., et al.,
 PLOS One.

Endorsing Complexity Through Diversity: Computational Psychiatry Meets Big Data Analytics,
 Kopal, J. & Bzdok, D.,
 Biological Psychiatry, S0006-3223(22)01479-2.

Leadership, Mentoring & Skills

Grant contribution,
 Horizon Europe grants: environMENTAL, Generative AI 4 EU
 Marie Curie Postdoctoral Fellowship
 Barrande Fellowship – mobility grant between Czech Republic and France.

Community Building,

Design and lead half-day collaborative workshops for the Centre for Precision Psychiatry at UiO, bringing together the Andreassen and Westlye research groups every 3-4 months. Program includes research presentations, collaborative tasks, and strategic goal-setting to promote cross-team collaboration and grant development.

Organize social initiatives to foster team cohesion across the department.

Science communication,

Oral presentations at World Congress of Psychiatry, Nordic Conference on Future Health, World Congress of Psychiatric Genetics, and the Organization for Human Brain Mapping.

Data visualization consultant, University of Oslo.

Supervision & mentoring,

Master and Bachelor students in computational neuroscience and machine learning.

Other activities,

Reviewer for NeuroImage, Scientific Reports, OHBM conference

Member of senior researcher hiring committees: University of Oslo, Mila - Quebec AI Institute.

Technical Skills

Development	Python (scikit-learn, PyTorch, pandas, numpy), R, MATLAB, bash Git version control Docker/Singularity containerization HPC clusters (SLURM)
ML & Statistics	Supervised/unsupervised learning, Bayesian modeling (PyMC3), multivariate methods (PCA, CCA, PLS), time-frequency analysis, tensor decomposition (PARAFAC2)

International training courses

Nov 2015	Universidad Politecnica de Madrid, Madrid, Spain, <i>Athens exchange program: Biosignal processing.</i>
Apr 2014	Technische Universitat Munich, Munich, Germany, <i>Athens exchange program: Manipulation of time series in time and frequency spectrum.</i>
Nov 2013	Ecole Nationale Supérieure de Techniques Avancées, Paris, France, <i>Athens exchange program: Medical Imaging.</i>

Languages

Czech	Native Speaker	
English	Fluent (C1+)	+ 3 semesters of Academic writing course
French	Fluent (C1)	
Norwegian	B1, actively improving	

References available upon request.